GitLab — Zero to Master

21 topics, grouped as Beginner → Intermediate → Advanced → Master. Work through the topics in order; type every code example yourself and finish the practice tasks.

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Beginner Level

01. Introduction to GitLab

DEFINITION

GitLab is a DevOps platform: Git hosting, issues, merge requests, CI/CD, container registry and security scanning in one product (SaaS or self-managed).

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A complete school campus with classrooms, library and exam hall in one place; GitLab bundles code hosting, CI/CD and security.

தமிழ்

வகுப்பறைகள், நூலகம், தேர்வு அறை எல்லாம் ஒரே இடத்தில் உள்ள பள்ளி வளாகம் போல GitLab code hosting, CI/CD, security-ஐ இணைக்கிறது.

PROS

- All-in-one DevOps platform
- Strong built-in CI/CD
- Self-hosting option
- Good security scanning

CONS

- Heavy to self-host
- Some features only in paid tiers
- UI can feel crowded

ADVANCED / NEW FEATURES

- GitLab Duo AI assistance
- CI/CD Components and Catalog
- Pipeline inputs
- Runner autoscaling improvements

02. Accounts, Groups and Projects

DEFINITION

Create an account, groups (teams) and projects (repositories); set visibility and roles (Guest, Reporter, Developer, Maintainer, Owner).

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A company with departments (groups) and projects inside them; roles decide who can do what.

தமிழ்

துறைகளும் (groups) அவற்றுக்குள் projects உள்ள நிறுவனம்; roles யார் என்ன செய்யலாம் என்பதைத் தீர்மானிக்கும்.

PROS

- Clear organisation
- Fine-grained roles

CONS

- Permission model can be complex

ADVANCED / NEW FEATURES

- Custom roles
- Subgroups and namespaces

03. SSH Keys and Cloning

DEFINITION

Add an SSH key to your profile and clone with SSH or HTTPS.

SYNTAX

```
ssh-keygen -t ed25519
```

EXAMPLE

```
ssh-keygen -t ed25519
git clone git@gitlab.com:group/project.git
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A personal key for your office door instead of typing a password every time; SSH keys log you in securely.

தமிழ்

ஒவ்வொரு முறையும் கடவுச்சொல் அடிக்காமல் அலுவலகக் கதவுக்கான தனிச் சாவி போல SSH keys பாதுகாப்பாக உள்நுழைய உதவுகின்றன.

PROS

- Secure and convenient
- No password prompts

CONS

- Key management needed

ADVANCED / NEW FEATURES

- ed25519 keys
- Signed commits with SSH keys

04. Issues, Labels and Milestones

DEFINITION

Track tasks and bugs; use labels, boards, milestones and weights.

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A complaint register and task board for the whole office; issues and boards track work.

தமிழ்

முழு அலுவலகத்துக்கான புகார் பதிவேடு, பணிப் பலகை போல issues, boards பணிகளைக் கண்காணிக்கின்றன.

PROS

- Planning and tracking
- Labels and milestones

CONS

- Advanced portfolio features are paid

ADVANCED / NEW FEATURES

- Work items and epics
- Issue boards

05. Merge Requests

DEFINITION

Propose changes from a branch, request review, discuss and merge.

EXAMPLE

```
git push -o merge_request.create origin feature/x
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Submitting your assignment for the teacher to review before it is added to the final book; a merge request does that for code.

தமிழ்

இறுதிப் புத்தகத்தில் சேர்க்கும் முன் ஆசிரியர் பார்வையிட பணியைச் சமர்ப்பிப்பது போல merge request குறியீட்டுக்கு அதைச் செய்கிறது.

PROS

- Review and discussion
- CI results shown

CONS

- Large MRs are hard to review

ADVANCED / NEW FEATURES

- Merge trains
- AI-assisted summaries (Duo)

06. Web IDE and Wiki

DEFINITION

Edit files in the browser; document with the project wiki.

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Editing a notice on the office board directly with a pen instead of going home to rewrite it; the Web IDE edits files in the browser.

தமிழ்

வீட்டுக்குச் சென்று மீண்டும் எழுதாமல் அலுவலகப் பலகையில் நேரடியாகத் திருத்துவது போல Web IDE browser-இலேயே கோப்புகளைத் திருத்துகிறது.

PROS

- No local setup
- Wiki for docs

CONS

- Limited compared with full IDEs

ADVANCED / NEW FEATURES

- Web IDE based on VS Code
- Workspaces

Intermediate Level

07. Branch Protection and Approvals

DEFINITION

Protect `main`, require approvals, code owners and pipelines to pass before merge.

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Only the principal can change the main notice board and needs two teachers' approval; protected branches enforce rules.

தமிழ்

முதன்மைப் பலகையை முதல்வர் மட்டுமே மாற்ற முடியும், இரு ஆசிரியர் ஒப்புதல் வேண்டும்; protected branches விதிகளை நடைமுறைப்படுத்துகின்றன.

PROS

- Safer main branch
- Required approvals and CODEOWNERS

CONS

- Can slow urgent fixes

ADVANCED / NEW FEATURES

- Push rules
- Approval policies

08. Introduction to CI/CD and .gitlab-ci.yml

DEFINITION

Pipelines made of stages and jobs defined in `.gitlab-ci.yml`; runners execute jobs.

SYNTAX

```
stages: [build, test]
build:
  script: [npm ci, npm run build]
```

EXAMPLE

```
stages: [build, test]
build-job:
  stage: build
  script:
    - npm ci
    - npm run build
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A factory assembly line: build, inspect, pack and dispatch happen automatically for every change.

தமிழ்

தொழிற்சாலை அசெம்பிளி வரிசை: ஒவ்வொரு மாற்றத்துக்கும் build, சோதனை, பேக்கிங், அனுப்புதல் தானாக நடக்கும்.

PROS

- Automation with simple YAML
- Integrated with MRs

CONS

- YAML can get long
- Runner costs

ADVANCED / NEW FEATURES

- CI/CD Components
- Pipeline inputs
- Steps keyword

09. Runners and Executors

DEFINITION

Shared vs project runners; Docker, shell and Kubernetes executors; register a runner.

SYNTAX

```
gitlab-runner register
```

EXAMPLE

```
gitlab-runner register
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Workers in the factory who actually do the jobs; runners execute pipeline jobs on machines or containers.

தமிழ்

தொழிற்சாலையில் உண்மையில் வேலை செய்யும் தொழிலாளர்கள் போல runners pipeline jobs-ஐ இயந்திரங்களில் அல்லது containers-இல் இயக்குகின்றன.

PROS

- Flexible executors (Docker, K8s)
- Autoscaling

CONS

- Self-managed runners need maintenance

ADVANCED / NEW FEATURES

- Runner fleet autoscaling
- Instance runners

10. Variables and Secrets

DEFINITION

CI/CD variables (masked, protected, file); predefined variables like `$CI_COMMIT_SHA`.

SYNTAX

```
variables:
  KEY: value
```

EXAMPLE

```
script:
  - echo $CI_COMMIT_BRANCH
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Keeping the godown key with the manager rather than writing it on the wall; masked variables keep secrets out of logs.

தமிழ்

கிடங்குச் சாவியை சுவரில் எழுதாமல் மேலாளரிடம் வைப்பது போல masked variables ரகசியங்களை log-இல் வராமல் காக்கின்றன.

PROS

- Secure config
- Environment scopes

CONS

- Mis-scoped variables leak secrets

ADVANCED / NEW FEATURES

- OIDC ID tokens for cloud auth
- External secrets integration

11. Artifacts and Caching

DEFINITION

Pass build outputs between jobs and speed up with caches.

SYNTAX

```
artifacts:
  paths: [dist/]
cache:
  paths: [node_modules/]
```

EXAMPLE

```
artifacts:
  paths: [dist/]
cache:
  paths: [node_modules/]
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Handing the finished boxes to the next station (artifacts) and keeping common tools ready (cache).

தமிழ்

முடிந்த பெட்டிகளை அடுத்த நிலையத்துக்குக் கொடுப்பது (artifacts), பொதுவான கருவிகளை தயாராக வைப்பது (cache) போல.

PROS

- Faster pipelines
- Job to job file sharing

CONS

- Cache invalidation is tricky

ADVANCED / NEW FEATURES

- Cache keys with files
- Artifact expiry policies

12. Environments and Deployments

DEFINITION

Deploy jobs with environments, manual approvals and review apps.

SYNTAX

```
environment: production
when: manual
```

EXAMPLE

```
deploy:
  stage: deploy
  environment: production
  when: manual
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A shop opening rule: staging is the trial counter and production is the real counter, with the owner's approval before opening; environments model that.

தமிழ்

முயற்சி கவுண்டர் (staging), உண்மை கவுண்டர் (production), திறக்கும் முன் உரிமையாளர் ஒப்புதல் போல environments அதை மாதிரியாக்குகின்றன.

PROS

- Deployment tracking
- Manual approvals
- Rollback

CONS

- Config effort

ADVANCED / NEW FEATURES

- Protected environments
- Deployment approvals

Advanced Level

13. Advanced Pipelines

DEFINITION

`rules`, `needs` (DAG), `include`, `extends`, templates, parent-child and multi-project pipelines.

SYNTAX

```
needs: [build]
rules:
  - if: $CI_COMMIT_BRANCH == "main"
```

EXAMPLE

```
test:
  needs: [build]
  rules:
    - if: $CI_COMMIT_BRANCH == "main"
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A kitchen with parallel stations that start as soon as their ingredients are ready, not waiting for everyone; DAG pipelines run faster.

தமிழ்

பொருட்கள் தயாரானவுடன் காத்திருக்காமல் தொடங்கும் இணை நிலையங்கள் உள்ள சமையலறை போல DAG pipelines வேகமாக இயங்குகின்றன.

PROS

- Faster, flexible pipelines
- Reusable templates

CONS

- Complex configs are hard to debug

ADVANCED / NEW FEATURES

- `rules`, `needs`, `include`, `extends`
- Parent-child pipelines
- Components

14. Container Registry and Docker Builds

DEFINITION

Build and push images to the GitLab registry in CI.

SYNTAX

```
docker build -t $CI_REGISTRY_IMAGE .
```

EXAMPLE

```
script:
  - docker build -t $CI_REGISTRY_IMAGE:$CI_COMMIT_SHA .
  - docker push $CI_REGISTRY_IMAGE:$CI_COMMIT_SHA
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A built-in store room for ready packed containers; the registry stores your images next to your code.

தமிழ்

தயாரான பேக் செய்த containers-க்கான உள்ளமைந்த கிடங்கு போல registry உங்கள் images-ஐ code அருகே சேமிக்கிறது.

PROS

- Integrated registry
- Cleanup policies

CONS

- Storage costs

ADVANCED / NEW FEATURES

- Dependency proxy
- Protected container tags

15. Security Scanning

DEFINITION

SAST, dependency, secret and container scanning, DAST via templates.

SYNTAX

```
include:
  - template: Security/SAST.gitlab-ci.yml
```

EXAMPLE

```
include:
  - template: Security/SAST.gitlab-ci.yml
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Security guards who check every parcel automatically; scanners find vulnerabilities and leaked secrets.

தமிழ்

ஒவ்வொரு பார்சலையும் தானாகச் சோதிக்கும் காவலர்கள் போல scanners பாதிப்புகளையும் கசிந்த ரகசியங்களையும் கண்டுபிடிக்கின்றன.

PROS

- Shift-left security
- Results in MRs

CONS

- Advanced features are paid
- False positives

ADVANCED / NEW FEATURES

- Secret push protection
- Dependency scanning with SBOM
- AI vulnerability explanations

16. Package Registry and Releases

DEFINITION

Publish npm, Maven, PyPI packages; create releases with notes and assets.

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A shop that shares its own price list publicly and keeps tagged versions of products; package registry and releases manage versions.

தமிழ்

தன் விலைப்பட்டியலைப் பொதுவில் பகிர்ந்து பதிப்புகளை குறிப்பிட்டு வைக்கும் கடை போல package registry, releases பதிப்புகளை நிர்வகிக்கின்றன.

PROS

- Private packages
- Release notes automation

CONS

- Per-ecosystem differences

ADVANCED / NEW FEATURES

- CI/CD Catalog publishing
- Release evidence

17. GitLab Pages

DEFINITION

Host static sites from a `pages` job.

SYNTAX

```
pages:
  script: [...]
  artifacts: { paths: [public] }
```

EXAMPLE

```
pages:
  script: [mkdir public, cp -r dist/* public]
  artifacts: { paths: [public] }
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A free notice board outside the office to display your website; GitLab Pages hosts static sites.

தமிழ்

உங்கள் வலைத்தளத்தைக் காட்ட அலுவலகத்துக்கு வெளியே இலவசப் பலகை போல GitLab Pages static தளங்களை host செய்கிறது.

PROS

- Free static hosting
- Custom domains

CONS

- Static only

ADVANCED / NEW FEATURES

- Parallel deployments
- Pages with CI

18. Kubernetes Integration and GitOps

DEFINITION

Deploy to clusters using the GitLab agent for Kubernetes and GitOps flows.

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

An automatic stock manager who matches the shelf with the register in Git; GitOps keeps clusters equal to what is written in Git.

தமிழ்

அலமாரியை Git பதிவேட்டுடன் பொருத்தும் தானியங்கி இருப்பு மேலாளர் போல GitOps clusters-ஐ Git-இல் எழுதியதற்கு சமமாக வைக்கிறது.

PROS

- Auditable changes
- Self-healing clusters

CONS

- Cluster agent setup

ADVANCED / NEW FEATURES

- GitLab agent for Kubernetes
- Flux integration

Master Level

19. Self-managed GitLab Administration

DEFINITION

Install via Omnibus/Helm, backups, upgrades, LDAP/SSO, runners at scale, monitoring.

SYNTAX

```
sudo gitlab-backup create
```

EXAMPLE

```
sudo gitlab-backup create
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Running your own school building means you handle repairs and backups yourself; self-managed GitLab needs upgrades, backups and monitoring.

தமிழ்

சொந்த பள்ளிக் கட்டிடம் நடத்தினால் பழுது, backup-ஐ நீங்களே பார்க்க வேண்டும்; self-managed GitLab-க்கு upgrades, backups, monitoring தேவை.

PROS

- Full control and data residency
- Customisable

CONS

- Operational effort
- Resource hungry

ADVANCED / NEW FEATURES

- GitLab Dedicated
- Cells architecture for scale

20. Pipeline Optimisation and Governance

DEFINITION

Fast pipelines, compliance pipelines, merge trains, protected environments, audit events.

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Reducing waiting time at each counter and checking that every rule is followed; pipeline optimisation and governance keep CI fast and compliant.

தமிழ்

ஒவ்வொரு கவுண்டரிலும் காத்திருப்பைக் குறைத்து அனைத்து விதிகளும் பின்பற்றப்படுகிறதா என்று பார்ப்பது போல pipeline optimisation, governance CI-ஐ வேகமாகவும் இணக்கமாகவும் வைக்கின்றன.

PROS

- Lower costs
- Compliance evidence

CONS

- Needs ongoing tuning

ADVANCED / NEW FEATURES

- Merge trains
- Compliance pipelines
- Audit events streaming

21. Real Project: Full CI/CD Pipeline

DEFINITION

Build -> test -> scan -> containerise -> deploy a web app to staging and production with approvals.

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A full factory line from raw material to delivery: build, test, scan, containerise and deploy in one pipeline.

தமிழ்

மூலப்பொருள் முதல் விநியோகம் வரை முழு தொழிற்சாலை வரிசை: build, test, scan, containerise, deploy எல்லாம் ஒரே pipeline-இல்.

PROS

- End-to-end DevOps practice

CONS

- Takes time to design

ADVANCED / NEW FEATURES

- Use CI/CD Components and OIDC deployments

Interview Question Bank

Q1. GitLab vs GitHub?
Both host Git; GitLab bundles a complete DevOps platform with built-in CI/CD, registry and security, and can be self-hosted.

Q2. What is a GitLab Runner?
An agent that executes CI/CD jobs on a machine/container.

Q3. What is `.gitlab-ci.yml`?
The file defining pipeline stages, jobs and rules.

Q4. Stage vs job?
A job is a unit of work; stages group jobs that run in order, jobs in a stage run in parallel.

Q5. Artifacts vs cache?
Artifacts are job outputs passed on; cache stores dependencies to speed up repeated jobs.

Q6. What is a merge request?
A request to merge a source branch into a target with review and CI checks.

Q7. How do you keep secrets safe?
Masked/protected CI variables or external secret managers; never commit them.

Q8. What are protected branches?
Branches with restricted push/merge rights.

Q9. What is `rules:` vs `only/except`?
`rules` is the modern, more flexible way to control when jobs run.

Q10. What is GitLab Pages?
Static site hosting directly from a repository.

Capstone Project Ideas

1. CI pipeline for a Node app
2. Docker build and push to registry
3. Deploy to a VPS with environments
4. Publish a static site with Pages
5. Security scanning pipeline